IS1111 Tutorial 7 – Importing from Modules/Packages & Dictionaries cont.

1) **Dictionary Counter in a Module**
   Create two files: main.py and drink_counter.py.
   Place following list in main.py:

```
drinks = ["Latte", "Americano", "Latte", "Espresso", "Latte", "Americano"]
```

   a. **Inside drink_counter.py, write a function called count_drinks(drinks)**
      The function must:
      - **take a list as a parameter**
      - **return a dictionary**
      - **key = drink name**
      - **value = number of times it appears**

   b. **In main.py import ONLY the function**
      Call the function using the drinks list and print the returned dictionary
   c. **In main.py import the WHOLE module**
      Call the function using dot notation
      Print the result

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2) **Module with Dictionary + Top-Level Code**
   Create two files: main.py and user_logins.py

   a. **In user_logins.py, create this dictionary**

```
logins = {"Katie": 1, "Ben": 2, "Ciara": 1}
```

   b. **Add a top-level print statement**

```
print("Loading user_logins...")
```

   c. **Write a function:**
      - **add_login(logins, username)**
      - **If the username exists → increase the count**
      - **If it does not exist → create it with value 1**
   d. **In main.py, import user_logins**
   e. **Call:**
      - **add_login for "Ben"**
      - **add_login for "David"**
   f. **Print the updated dictionary**

Run main.py and observe what prints during import

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3) if__name__ == "__main__" Guard + Safe Lookup
Modify user_logins.py

a. **Write a function:**
   - **get_login_count(logins, username)**
   - **Return the login count if the user exists**
   - **Return 0 if the user does not exist**

b. **Add a test block at the bottom of users_logins.py**

```
if __name__ == "__main__":
```

Inside the block:
   - **Print Katie's login count**
   - **Print David's login count**
   - **Call add_login(logins, "Katie")**
   - **Print the updated dictionary**

c. **Run user_logins.py directly.**
   Then run main.py.
   Observe the difference in output.

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4) Nested Dictionary + Separation of Concerns
Create three files: main.py, employee_Data.py, and employee_utils.py

a. **In employee_data.py, store ONLY this nested dictionary:**
   - **Employee ID as key**
   - **Value is a dictionary with name, dept, salary**
     (Use 4 employees: E001-E004)

b. **In employee_utils.py, write:**
   - **get_salary(employees, emp_id)**
     - **Return salary if employee exists**
     - **Return None otherwise**

   - **increase_salary(employees, emp_id, amount)**
     - **Increase salary if employee exists**
     - **Return True if updated**
     - **Return False otherwise**

c. **In main.py, import:**
   - **employees from employee_data.py**
   - **both functions from employee_utils.py**

d. **Print the salary for "E002"**
e. **Print the salary for "E999" (safe lookup)**
f. **Increase Ben's salary (E004) by 3000**
g. **Print Ben's updated salary**